

Linear algebra recap

Motivation: linear model

We would like to discuss the linear regression model, given as

$$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$$

where $\mathbf{y} \in \mathbb{R}^n$, $X \in \mathbb{R}^{n \times m}$, $\boldsymbol{\beta} \in \mathbb{R}^m$, $\boldsymbol{\epsilon} \in \mathbb{R}^n$.

Matrix operations

Matrix multiplication

Let $A \in \mathbb{R}^{n \times p}$, $B \in \mathbb{R}^{p \times m}$.

We will use indices to denote individual elements of a matrix: A_{ij} denotes the element in the i -th row and j -th column of the matrix A .

We define the matrix product $C = AB$ as $C \in \mathbb{R}^{n \times m}$, where

$$C_{ij} = \sum_{k=1}^p A_{ik}B_{kj}.$$

Note that:

- Matrix multiplication is defined only if the number of columns of A equals the number of rows of B .
- Matrix multiplication is not commutative, i.e. generally $AB \neq BA$.

Identity matrix

The identity matrix $I \in \mathbb{R}^{n \times n}$ is a diagonal matrix with ones on the diagonal (and zeros elsewhere). For example, for $n = 3$:

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

For any $A \in \mathbb{R}^{n \times n}$, we have

$$AI = IA = A.$$

Inverse matrix

For $A \in \mathbb{R}^{n \times n}$, the inverse matrix A^{-1} is defined (if it exists) such that

$$AA^{-1} = A^{-1}A = I.$$

The inverse does not exist for all matrices. If A^{-1} exists, we say that A is **invertible** (or **regular**); otherwise, A is called **singular**.

Transpose

For $A \in \mathbb{R}^{n \times p}$, the transpose is defined as $A^T \in \mathbb{R}^{p \times n}$ such that

$$(A^T)_{ij} = A_{ji}.$$

That is, rows of A become columns of A^T , and columns of A become rows of A^T .

Properties:

- $(A^T)^T = A$
- $(AB)^T = B^T A^T$

A matrix $A \in \mathbb{R}^{n \times n}$ is called **symmetric** if $A^T = A$.

Derivatives with respect to vectors

Let $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, $A \in \mathbb{R}^{n \times n}$.

- $\frac{\partial}{\partial \mathbf{b}} (\mathbf{a}^T \mathbf{b}) = \frac{\partial}{\partial \mathbf{b}} (\mathbf{b}^T \mathbf{a}) = \mathbf{a}$
- $\frac{\partial}{\partial \mathbf{b}} (\mathbf{b}^T A \mathbf{b}) = (A + A^T) \mathbf{b}$
- If A is symmetric ($A^T = A$), then

$$\frac{\partial}{\partial \mathbf{b}} (\mathbf{b}^T A \mathbf{b}) = 2A \mathbf{b}$$

Linear model - solving for β

- Loss function (obtained from negative log-likelihood) is

$$\begin{aligned} \epsilon^T \epsilon &= (\mathbf{y} - X\beta)^T (\mathbf{y} - X\beta) \\ &= \mathbf{y}^T \mathbf{y} - \mathbf{y}^T X\beta - \beta^T X^T \mathbf{y} + \beta^T X^T X\beta \\ &= \mathbf{y}^T \mathbf{y} - 2\beta^T X^T \mathbf{y} + \beta^T X^T X\beta \end{aligned}$$

- Setting the derivative (with respect to β) to 0:

$$\frac{\partial \epsilon^T \epsilon}{\partial \beta} = -2X^T \mathbf{y} + 2X^T X\beta = 0$$

- From that, we have

$$X^T X \boldsymbol{\beta} = X^T \mathbf{y}$$

- Multiplying from the left by $(X^T X)^{-1}$ gives

$$\hat{\boldsymbol{\beta}} = (X^T X)^{-1} X^T \mathbf{y}$$

Note: $(X^T X)^{-1}$ may not exist!

Additional resources

- [Essence of linear algebra](#) - YouTube series by 3Blue1Brown.
- Chapter 2 of [Goodfellow's Deep Learning textbook](#).